

Hydrogen bonding

A hydrogen bond is the attractive interaction of an electropositive hydrogen atom (which is covalently bonded to an electronegative atom e.g. N, O or F) with an electronegative atom with a lone pair like N, O or F etc. The atom which is covalently bonded to the hydrogen is called hydrogen bond donor, and the part which offers a lone pair is called hydrogen bond acceptor. Most of the hydrogen bonds A-H...D are asymmetric, even if A and D are the same elements.

The hydrogen bond (5 to 30 kJ/mol) is stronger than a van der Waals interaction, but weaker than covalent or ionic bonds. This type of bond occurs in both inorganic molecules such as water and organic molecules such as DNA.

Liquids that display hydrogen bonding are called associated liquids.

ΔH°_f (bond enthalpy) values of hydrogen bonds depend on the nature of donor and acceptor and are affected by pressure and temperature.

Examples:

F-H...F ($\Delta H^\circ_f = 29$ kJ/mol): An electronegative atom in the donor part and the same atom as acceptor. Liquid hydrogen fluoride forms zigzag shaped associates of the linear HF molecules.

H-O-H...OH₂ (21 kJ/mol) and O-H...N (29 kJ/mol): Although the N and O atoms have similar properties (mass, electronegativity, size etc.) the hydrogen bond enthalpy differs slightly because one O atom acts as a donor as well as an acceptor for two H bonds, in contrast to N or F (see discussion below).

N-H...O (8 kJ/mol) and N-H...N (13 kJ/mol): We have a similar discussion as above. The O atom acts as an acceptor for two hydrogen bonds.

Consequences of H-bonds on properties:

Water has a very high boiling point, melting point and viscosity compared to other similar liquids which do not form such strong hydrogen bonds. Water is unique because its oxygen atom has two lone pairs and two hydrogen atoms. The total number of bonds of a water molecule is four. Hydrogen fluoride, which has three lone pairs on the F atom but one H atom, can form only two bonds. For ammonia it is the opposite case: Three hydrogen atoms but only one lone pair on the N atom.

By comparing H₂O with H₂S, H₂Se and H₂Te, we can see that the boiling points will increase from H₂S to H₂Te because of the increasing molecular weights. That means H₂O is supposed to have a lower boiling point than H₂S, but the opposite can be observed and thus be explained by the formation of strong hydrogen bonds.

Hydrogen bond lengths can be characterized by cross-correlated relaxation method in NMR technique, vibrational spectroscopy (also with isotopically diluted systems), neutron diffraction as well as quantum chemical calculations.

Literature ???